Peter J. Marshall

Curriculum Vitae

Professor
Department of Psychology & Neuroscience
Temple University
Weiss Hall
1701 North 13th St.
Philadelphia, PA 19122

peter.marshall@temple.edu
peterjmarshall.com

Professional Appointments

2015–	Professor, Department of Psychology & Neuroscience Temple University (formerly Department of Psychology; name change effective 01/22)
2016–2026	Chair, Department of Psychology & Neuroscience Temple University
2014–2015	Visiting Scholar Institute for Learning & Brain Sciences, University of Washington
2010–2015	Associate Professor Department of Psychology, Temple University
2004–2010	Assistant Professor Department of Psychology, Temple University
1999–2004	Research Assistant Professor Department of Human Development, University of Maryland
1997–1999	Postdoctoral Research Associate Department of Human Development, University of Maryland

Education

1997	Ph.D., University of Cambridge Area of Research: Developmental Psychology Advisor: Joan Stevenson-Hinde, Sc.D.
1993	B.A. Hons. (Cantab) First Class, University of Cambridge Natural Sciences (Part II Zoology)

Publications

Peer-Reviewed Articles

Marshall, P. J. (in press). Psychology in 2125: A developmental science of living sense-making. *Perspectives on Psychological Science*. https://doi.org/10.1177/17456916261477319. Preprint available at https://doi.org/10.31234/osf.io/d7y3n_v3

Campbell, K., Horger, M. N., Pavlakos, G., Simonetti, V., Bulgheroni, M., Parma, V., & Marshall, P. J. (in press). Longitudinal infant prehension development: Novel insights from the moment of first contact. *Developmental Psychobiology*.

Zweben, D., Zaneski, L., Chiu, M., Green, H., Brooks-Justice, J., Dachs, A., Vince, M. A., Olino, T. M., Marshall, P., Steinberg, L., & Chein, J. M. (2026). Clarifying the association between digital media engagement and risk taking. *Technology in Society, 88*, 103491. https://doi.org/10.1016/j.techsoc.2026.103491

Campbell, K., Horger, M. N., Pavlakos, G., Parma, V. & Marshall, P. J. (2026). Dynamics of infant preshaping: Longitudinal analysis using automated hand pose tracking from home video. *Infant Behavior and Development*, *84*, 102232. https://doi.org/10.1016/j.infbeh.2026.102232

Langer, A., Martinez, S., Marshall, P. J., & Chein, J. (2026). Children's susceptibility to content generated by artificial intelligence. *Technology in Society, 86*, 103303. https://doi.org/10.1016/j.techsoc.2026.103303

Langer, A., Wilson, J. R., Howard, L., & Marshall, P. J. (2026). The influence of individual characteristics on children's learning with a social robot versus a human. *Scientific Reports, 16*, 3488. https://doi.org/10.1038/s41598-025-27476-x

Wilson, J. R., Langer, A., Howard, L., & Marshall, P. J. (2025). Age-related differences in children's spontaneous gesturing with a robot versus human instructor. In *2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)* (pp. 43–50). IEEE. https://doi.org/10.1109/RO-MAN63969.2025.11217556

Marshall, P. J. (2024). Towards a biologically coherent account of the brain and how it develops. *Human Development*, *68*(5–6), 209–220. https://doi.org/10.1159/000540024

Horger, M. N., Campbell, K., & Marshall, P. J. (2024). Electrophysiological responses to digit stimulation in a tactile oddball paradigm. *International Journal of Psychophysiology, 203*, Article 112391. https://doi.org/10.1016/j.ijpsycho.2024.112391

Yang, Y., Langer, A., Howard, L., Marshall, P. J., & Wilson, J. R. (2024). Towards an ontology for generating behaviors for socially assistive robots helping young children. *Proceedings of the AAAI Symposium Series, 2*(1), 213–218. https://doi.org/10.1609/aaaiss.v2i1.27674

Taylor, J., Weiss, S. M., & Marshall, P. J. (2023). Genes, genomes, and developmental process. *Behavioral and Brain Sciences, 46*, Article e204. https://doi.org/10.1017/S0140525X22002199

Langer, A., Marshall, P. J., & Levy-Tzedek, S. (2023). Ethical considerations in child-robot interactions. *Neuroscience & Biobehavioral Reviews, 151*, Article 105230. https://doi.org/10.1016/j.neubiorev.2023.105230

Ma, L., Marshall, P. J., & Wright, W. G. (2023). The order of attentional focus instructions affects how postural control processes compensate for multisensory mismatch: A crossover study. *Experimental Brain Research, 241*, 1393–1409. https://doi.org/10.1007/s00221-023-06610-z

Weiss, S. M., & Marshall, P. J. (2023). Anticipation across modalities in children and adults: Relating anticipatory alpha rhythm lateralization, reaction time, and executive function. *Developmental Science, 26*, e13277. https://doi.org/10.1111/desc.13277

Ma, L., Marshall, P. J., & Wright, W. G. (2022). The impact of external and internal focus of attention on visual dependence and EEG alpha oscillations during postural control. *Journal of NeuroEngineering and Rehabilitation, 19*, 81. https://doi.org/10.1186/s12984-022-01059-7

Shen, G., Weiss, S. M., Meltzoff, A. N., Allison, O. N., & Marshall, P. J. (2022). Exploring developmental changes in infant anticipation and perceptual processing: EEG responses to tactile stimulation. *Infancy, 27*, 97–114. https://doi.org/10.1111/infa.12438

Marshall, P. J., Houser, T. M., & Weiss, S. M. (2021). The shared origins of embodiment and development. *Frontiers in Systems Neuroscience, 15*, 726403. https://doi.org/10.3389/fnsys.2021.726403

Belhassein, K., Marshall, P. J., Badets, A., & Bouquet, C. A. (2021). Social learning of action-effect associations: Modulation of action control following observation of virtual action's effects. *Attention, Perception, & Psychophysics, 83*, 484–496. https://doi.org/10.3758/s13414-020-02157-1

Taylor, J., Weiss, S. M., & Marshall, P. J. (2020). "Alexa, how are you feeling today?" Mind perception, smart speakers, and uncanniness. *Interaction Studies, 21*, 329–352. https://doi.org/10.1075/is.19015.tay

Meltzoff, A. N., & Marshall, P. J. (2020). Importance of body representations in social-cognitive development: New insights from infant brain science. *Progress in Brain Research, 254*, 25–48. https://doi.org/10.1016/bs.pbr.2020.07.009

Weiss, S. M., Laconi, R. N., & Marshall, P. J. (2020). Individual differences in anticipatory mu rhythm modulation are associated with executive function and processing speed. *Cognitive, Affective, & Behavioral Neuroscience, 20*, 901–916. https://doi.org/10.3758/s13415-020-00809-1

Shen, G., Meltzoff, A. N., Weiss, S. M., & Marshall, P. J. (2020). Body representation in infants: Categorical boundaries of body parts as assessed by somatosensory mismatch negativity. *Developmental Cognitive Neuroscience*, *44*, 100795. https://doi.org/10.1016/j.dcn.2020.100795

Marshall, P. J., & Meltzoff, A. N. (2020). Body maps in the infant brain: Implications for neurodevelopmental disabilities. *Developmental Medicine & Child Neurology*, *62*, 778–783. https://doi.org/10.1111/dmcn.14540

Shen, G., Meltzoff, A. N., & Marshall, P. J. (2019). Body representations as indexed by oscillatory EEG activities in the context of tactile novelty processing. *Neuropsychologia*, *132*, 107144. https://doi.org/10.1016/j.neuropsychologia.2019.107144

Meltzoff, A. N., Saby, J. N., & Marshall, P. J. (2019). Neural representations of the body in 60-day-old human infants. *Developmental Science*, *22*, e12698. https://doi.org/10.1111/desc.12698

Drew, A. R., Meltzoff, A. N., & Marshall, P. J. (2018). Interpersonal influences on body representations in the infant brain. *Frontiers in Psychology*, *9*, 2601. https://doi.org/10.3389/fpsyg.2018.02601

Weiss, S. M., Meltzoff, A. N., & Marshall, P. J. (2018). Neural measures of anticipatory bodily attention in children: Relations with executive function. *Developmental Cognitive Neuroscience*, *34*, 148–158. https://doi.org/10.1016/j.dcn.2018.08.002

Smyk, N. J., Weiss, S. M., & Marshall, P. J. (2018). Sensorimotor oscillations during a reciprocal touch paradigm with a human or robot partner. *Frontiers in Psychology*, *9*, 2280. https://doi.org/10.3389/fpsyg.2018.02280

Shen, G., Weiss, S. M., Meltzoff, A. N., & Marshall, P. J. (2018). The somatosensory mismatch negativity as a window into body representations in infancy. *International Journal of Psychophysiology*, *134*, 144–150. https://doi.org/10.1016/j.ijpsycho.2018.10.013

Shen, G., Smyk, N. J., Meltzoff, A. N., & Marshall, P. J. (2018). Neuropsychology of human body parts: Exploring categorical boundaries of tactile perception using somatosensory mismatch responses. *Journal of Cognitive Neuroscience*, *30*, 1858–1869. https://doi.org/10.1162/jocn_a_01313

Bunlon, F., Gazeau, J.-P., Colloud, F., Marshall, P. J., & Bouquet, C. A. (2018). Joint action with a virtual robotic vs. human agent. *Cognitive Systems Research*, *52*, 816–827. https://doi.org/10.1016/j.cogsys.2018.09.017

Meltzoff, A. N., & Marshall, P. J. (2018). Human infant imitation as a social survival circuit. *Current Opinion in Behavioral Sciences*, *24*, 130–136. https://doi.org/10.1016/j.cobeha.2018.09.006

Meltzoff, A. N., Ramirez, R. R., Saby, J. N., Larson, E., Taulu, S., & Marshall, P. J. (2018). Infant brain responses to felt and observed touch of hands and feet: An MEG study. *Developmental Science*, *21*, e12651. https://doi.org/10.1111/desc.12651

Witherington, D. C., Overton, W. F., Lickliter, R., Marshall, P. J., & Narvaez, D. (2018). Metatheory and the primacy of conceptual analysis in developmental science. *Human Development*, *61*, 181–198. https://doi.org/10.1159/000490160

Shen, G., Meltzoff, A. N., & Marshall, P. J. (2018). Touching lips and hearing fingers: Effector-specific congruency between tactile and auditory stimulation modulates N1 amplitude and alpha desynchronization. *Experimental Brain Research*, *236*, 13–29. https://doi.org/10.1007/s00221-017-5104-3

Shen, G., Smyk, N. J., Meltzoff, A. N., & Marshall, P. J. (2018). Using somatosensory mismatch responses as a window into somatotopic processing of tactile stimulation. *Psychophysiology*, *55*, e13030. https://doi.org/10.1111/psyp.13030

Shen, G., Saby, J. N., Drew, A. R., & Marshall, P. J. (2017). Exploring potential social influences on brain potentials during anticipation of tactile stimulation. *Brain Research*, *1659*, 8–18. https://doi.org/10.1016/j.brainres.2017.01.022

Marshall, P. J. (2016). Embodiment and human development. *Child Development Perspectives*, *10*, 245–250. https://doi.org/10.1111/cdep.12190

Marshall, P. J., & Brenneman, K. (2016). Young children's developing understanding of the biological world. *Early Education and Development*, *27*, 1103–1108. https://doi.org/10.1080/10409289.2016.1220772

Saby, J. N., Meltzoff, A. N., & Marshall, P. J. (2016). Beyond the N1: A review of late somatosensory evoked responses in human infants. *International Journal of Psychophysiology*, *110*, 146–152. https://doi.org/10.1016/j.ijpsycho.2016.08.008

Marshall, P. J., & Meltzoff, A. N. (2015). Body maps in the infant brain. *Trends in Cognitive Sciences*, *19*, 499–505. https://doi.org/10.1016/j.tics.2015.06.012

Saby, J. N., Meltzoff, A. N., & Marshall, P. J. (2015). Neural body maps in human infants: Somatotopic responses to tactile stimulation in 7-month-olds. *NeuroImage*, *118*, 74–78. https://doi.org/10.1016/j.neuroimage.2015.05.097

Drew, A. R., Quandt, L. C., & Marshall, P. J. (2015). Visual influences on sensorimotor EEG responses during observation of hand actions. *Brain Research*, *1597*, 119–128. https://doi.org/10.1016/j.brainres.2014.11.048

Bunlon, F., Marshall, P. J., Quandt, L. C., & Bouquet, C. A. (2015). Influence of action-effect associations acquired by ideomotor learning on imitation. *PLOS ONE*, *10*, e0121617. https://doi.org/10.1371/journal.pone.0121617

Saby, J. N., Bouquet, C. A., & Marshall, P. J. (2014). Young children co-represent a partner's task: Evidence for a joint Simon effect in five-year-olds. *Cognitive Development*, *32*, 38–45. https://doi.org/10.1016/j.cogdev.2014.08.001

Marshall, P. J. (2014). Beyond different levels: Embodiment and the developmental system. *Frontiers in Psychology*, *5*, 929. https://doi.org/10.3389/fpsyg.2014.00929

Marshall, P. J., & Drew, A. R. (2014). What makes Simon Says so difficult for young children? *Journal of Experimental Child Psychology*, *126*, 112–119. https://doi.org/10.1016/j.jecp.2014.03.011

Marshall, P. J., & Meltzoff, A. N. (2014). Neural mirroring mechanisms and imitation in human infants. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *369*, 20130620. https://doi.org/10.1098/rstb.2013.0620

Quandt, L. C., & Marshall, P. J. (2014). The effect of action experience on sensorimotor EEG rhythms during action observation. *Neuropsychologia*, *56*, 401–408. https://doi.org/10.1016/j.neuropsychologia.2014.02.015

Marshall, P. J. (2013). Coping with complexity: Developmental systems and multilevel analyses in developmental psychopathology. *Development and Psychopathology*, *25*, 1311–1324. https://doi.org/10.1017/S0954579413000631

Marshall, P. J., Saby, J. N., & Meltzoff, A. N. (2013). Imitation and the developing social brain: Infants' somatotopic EEG patterns for acts of self and other. *International Journal of Psychological Research*, *6*, 22–29. https://doi.org/10.21500/20112084.714

Saby, J. N., Meltzoff, A. N., & Marshall, P. J. (2013). Infants' somatotopic neural responses to seeing human actions: I've got you under my skin. *PLOS ONE*, *8*, e77905. https://doi.org/10.1371/journal.pone.0077905

Marshall, P. J., Saby, J. N., & Meltzoff, A. N. (2013). Infant brain responses to object weight: Exploring goal-directed actions and self-experience. *Infancy*, *18*, 942–960. https://doi.org/10.1111/infa.12012

Quandt, L. C., Marshall, P. J., Bouquet, C. A., & Shipley, T. F. (2013). Somatosensory experiences with action modulate alpha and beta power during subsequent action observation. *Brain Research*, *1534*, 55–65. https://doi.org/10.1016/j.brainres.2013.08.043

Saby, J. N., Marshall, P. J., & Meltzoff, A. N. (2012). Neural correlates of being imitated: An EEG study in preverbal infants. *Social Neuroscience*, *7*, 650–661. https://doi.org/10.1080/17470919.2012.691429

Quandt, L. C., Marshall, P. J., Shipley, T. F., Beilock, S. L., & Goldin-Meadow, S. (2012). Sensitivity of alpha and beta oscillations to sensorimotor characteristics of action: An EEG study of action production and gesture observation. *Neuropsychologia*, *50*, 2745–2751. https://doi.org/10.1016/j.neuropsychologia.2012.08.005

Saby, J. N., & Marshall, P. J. (2012). The utility of EEG band power analysis in the study of infancy and early childhood. *Developmental Neuropsychology*, *37*, 253–273. https://doi.org/10.1080/87565641.2011.614663

Marshall, P. J., & Comalli, C. E. (2012). Young children's changing conceptualizations of brain function: Implications for teaching neuroscience in early elementary settings. *Early Education and Development*, *23*, 4–23. https://doi.org/10.1080/10409289.2011.616134

Saby, J. N., Marshall, P. J., Smythe, R., Bouquet, C. A., & Comalli, C. E. (2011). An investigation of the determinants of motor contagion in preschool children. *Acta Psychologica*, *138*, 231–236. https://doi.org/10.1016/j.actpsy.2011.06.008

Quandt, L. C., Marshall, P. J., Bouquet, C. A., Young, T., & Shipley, T. F. (2011). Experience with novel actions modulates frontal alpha EEG desynchronization. *Neuroscience Letters*, *499*, 37–41. https://doi.org/10.1016/j.neulet.2011.05.028

Bouquet, C. A., Shipley, T. F., Capa, R. L., & Marshall, P. J. (2011). Motor contagion: Goal-directed actions are more contagious than non-goal-directed actions. *Experimental Psychology*, *58*, 71–78. https://doi.org/10.1027/1618-3169/a000069

Capa, R. L., Marshall, P. J., Shipley, T. F., Salesse, R. N., & Bouquet, C. A. (2011). Does motor interference arise from mirror system activation? The effect of prior visuo-motor practice on automatic imitation. *Psychological Research*, *75*, 152–157. https://doi.org/10.1007/s00426-010-0303-6

Marshall, P. J., & Meltzoff, A. N. (2011). Neural mirroring systems: Exploring the EEG mu rhythm in infancy. *Developmental Cognitive Neuroscience*, *1*, 110–123. https://doi.org/10.1016/j.dcn.2010.09.001

Marshall, P. J., Young, T., & Meltzoff, A. N. (2011). Neural correlates of action observation and execution in 14-month-old infants: An event-related EEG desynchronization study. *Developmental Science*, *14*, 474–480. https://doi.org/10.1111/j.1467-7687.2010.00991.x

Marshall, P. J., Bouquet, C. A., Thomas, A. L., & Shipley, T. F. (2010). Motor contagion in young children: Exploring social influences on perception-action coupling. *Neural Networks*, *23*, 1017–1025. https://doi.org/10.1016/j.neunet.2010.07.007

McLaughlin, K. A., Fox, N. A., Zeanah, C. H., Sheridan, M. A., Marshall, P. J., & Nelson, C. A. (2010). Delayed maturation in brain electrical activity partially explains the association between early environmental deprivation and symptoms of attention-deficit/hyperactivity disorder. *Biological Psychiatry*, *68*, 329–336. https://doi.org/10.1016/j.biopsych.2010.04.005

Vanderwert, R. E., Marshall, P. J., Nelson, C. A., Zeanah, C. H., & Fox, N. A. (2010). Timing of intervention affects brain electrical activity in children exposed to severe psychosocial neglect. *PLOS ONE*, *5*, e11415. https://doi.org/10.1371/journal.pone.0011415

Drury, S. S., Theall, K. P., Smyke, A. T., Keats, B. J., Egger, H. L., Nelson, C. A., Fox, N. A., Marshall, P. J., & Zeanah, C. H. (2010). Modification of depression by COMT Val158Met polymorphism in children exposed to early severe psychosocial deprivation. *Child Abuse & Neglect*, *34*, 387–395. https://doi.org/10.1016/j.chiabu.2009.09.021

Marshall, P. J. (2010). The development of emotion. *WIREs Cognitive Science*, *1*, 417–425. https://doi.org/10.1002/wcs.51

Reeb-Sutherland, B. C., Vanderwert, R. E., Degnan, K. A., Marshall, P. J., Pérez-Edgar, K., Chronis-Tuscano, A., Pine, D. S., & Fox, N. A. (2009). Attention to novelty in behaviorally inhibited adolescents moderates risk for anxiety. *Journal of Child Psychology and Psychiatry*, *50*, 1365–1372. https://doi.org/10.1111/j.1469-7610.2009.02170.x

Marshall, P. J. (2009). Relating psychology and neuroscience: Taking up the challenges. *Perspectives on Psychological Science*, *4*, 113–125. https://doi.org/10.1111/j.1745-6924.2009.01111.x

Marshall, P. J., & Kenney, J. W. (2009). Biological perspectives on the effects of early psychosocial experience. *Developmental Review*, *29*, 96–119. https://doi.org/10.1016/j.dr.2009.05.001

Marshall, P. J., Bouquet, C. A., Shipley, T. F., & Young, T. (2009). Effects of brief imitative experience on EEG desynchronization during action observation. *Neuropsychologia*, *47*, 2100–2106. https://doi.org/10.1016/j.neuropsychologia.2009.03.022

Marshall, P. J., & Shipley, T. F. (2009). Event-related potentials to point-light displays of human actions in 5-month-old infants. *Developmental Neuropsychology*, *34*, 368–377. https://doi.org/10.1080/87565640902801866

Marshall, P. J., Reeb, B. C., & Fox, N. A. (2009). Electrophysiological responses to auditory novelty in temperamentally different 9-month-old infants. *Developmental Science*, *12*, 568–582. https://doi.org/10.1111/j.1467-7687.2008.00808.x

Fisher, K. R., Marshall, P. J., & Nanayakkara, A. R. (2009). Motivational orientation, error monitoring, and academic performance in middle childhood: A behavioral and electrophysiological investigation. *Mind, Brain, and Education*, *3*, 56–63. https://doi.org/10.1111/j.1751-228X.2008.01053.x

Ghera, M. M., Marshall, P. J., Fox, N. A., Zeanah, C. H., Nelson, C. A., Smyke, A. T., & Guthrie, D. (2009). The effects of a foster care intervention on socially deprived institutionalized children's attention and positive affect: Results from the BEIP study. *Journal of Child Psychology and Psychiatry*, *50*, 246–253. https://doi.org/10.1111/j.1469-7610.2008.01954.x

Zeanah, C. H., Egger, H. L., Smyke, A. T., Nelson, C. A., Fox, N. A., Marshall, P. J., & Guthrie, D. (2009). Institutional rearing and psychiatric disorders in Romanian preschool children. *American Journal of Psychiatry*, *166*, 777–785. https://doi.org/10.1176/appi.ajp.2009.08091438

Marshall, P. J., Reeb, B. C., & Fox, N. A. (2008). Effects of early intervention on EEG power and coherence in previously institutionalized children in Romania. *Development and Psychopathology*, *20*, 861–880. https://doi.org/10.1017/S0954579408000412

Hane, A. A., Fox, N. A., Henderson, H. A., & Marshall, P. J. (2008). Behavioral reactivity and approach-withdrawal bias in infancy. *Developmental Psychology*, *44*, 1491–1496. https://doi.org/10.1037/a0012855

Marx, B. P., Marshall, P. J., & Castro, F. (2008). The moderating effects of stimulus valence and arousal on memory suppression. *Emotion*, *8*, 199–207. https://doi.org/10.1037/1528-3542.8.2.199

Nelson, C. A., Zeanah, C. H., Fox, N. A., Marshall, P. J., Smyke, A. T., & Guthrie, D. (2007). Cognitive recovery in socially deprived young children: The Bucharest Early Intervention Project. *Science*, *318*, 1937–1940. https://doi.org/10.1126/science.1143921

Smyke, A. T., Koga, S. F., Johnson, D. E., Fox, N. A., Marshall, P. J., Nelson, C. A., Zeanah, C. H., & the BEIP Core Group. (2007). The caregiving context in institution-reared and family-reared infants and toddlers in Romania. *Journal of Child Psychology and Psychiatry*, *48*, 210–218. https://doi.org/10.1111/j.1469-7610.2006.01694.x

Marshall, P. J., & Fox, N. A. (2005). Relations between behavioral reactivity at 4 months and attachment classification at 14 months in a selected sample. *Infant Behavior and Development*, *28*, 492–502. https://doi.org/10.1016/j.infbeh.2005.06.002

Fox, N. A., Henderson, H. A., Marshall, P. J., Nichols, K. E., & Ghera, M. M. (2005). Behavioral inhibition: Linking biology and behavior within a developmental framework. *Annual Review of Psychology*, *56*, 235–262. https://doi.org/10.1146/annurev.psych.55.090902.141532

Marshall, P. J., Fox, N. A., & the BEIP Core Group. (2004). A comparison of the electroencephalogram between institutionalized and community children in Romania. *Journal of Cognitive Neuroscience*, *16*, 1327–1338. https://doi.org/10.1162/0898929042304723

Bar-Haim, Y., Fox, N. A., van Meenen, K. M., & Marshall, P. J. (2004). Children's narratives and patterns of cardiac reactivity. *Developmental Psychobiology*, *44*, 238–249. https://doi.org/10.1002/dev.20006

Henderson, H. A., Marshall, P. J., Fox, N. A., & Rubin, K. H. (2004). Psychophysiological and behavioral evidence for varying forms and functions of nonsocial behavior in preschoolers. *Child Development*, *75*, 251–263. https://doi.org/10.1111/j.1467-8624.2004.00667.x

Marshall, P. J., Bar-Haim, Y., & Fox, N. A. (2004). The development of P50 suppression in the auditory event-related potential. *International Journal of Psychophysiology*, *51*, 135–141. https://doi.org/10.1016/j.ijpsycho.2003.08.004

Zeanah, C. H., Nelson, C. A., Fox, N. A., Smyke, A. T., Marshall, P. J., Parker, S. W., & Koga, S. (2003). Designing research to study the effects of institutionalization on brain and behavioral development: The Bucharest Early Intervention Project. *Development and Psychopathology*, *15*, 885–907. https://doi.org/10.1017/S0954579403000452

Burgess, K. B., Marshall, P. J., Rubin, K. H., & Fox, N. A. (2003). Infant attachment and temperament as predictors of subsequent externalizing problems and cardiac physiology. *Journal of Child Psychology and Psychiatry*, *44*, 819–831. https://doi.org/10.1111/1469-7610.00167

Bar-Haim, Y., Marshall, P. J., Fox, N. A., Schorr, E., & Gordon-Salant, S. (2003). Mismatch negativity in socially withdrawn children. *Biological Psychiatry*, *54*, 17–24. https://doi.org/10.1016/S0006-3223(03)00175-6

Marshall, P. J., Bar-Haim, Y., & Fox, N. A. (2002). Development of the EEG from 5 months to 4 years of age. *Clinical Neurophysiology*, *113*, 1199–1208. https://doi.org/10.1016/S1388-2457(02)00163-3

Bar-Haim, Y., Marshall, P. J., & Fox, N. A. (2000). Developmental changes in heart period and high-frequency heart period variability from 4 months to 4 years of age. *Developmental Psychobiology*, *37*, 44–56. https://doi.org/10.1002/1098-2302(200007)37:1%3C44::AID-DEV6%3E3.0.CO;2-7

Marshall, P. J., Fox, N. A., & Henderson, H. A. (2000). Temperament as an organizer of development. *Infancy*, *1*, 239–244. https://doi.org/10.1207/S15327078IN0102_5

Stevenson-Hinde, J., & Marshall, P. J. (1999). Behavioral inhibition, heart period, and respiratory sinus arrhythmia: An attachment perspective. *Child Development*, *70*, 805–816. https://doi.org/10.1111/1467-8624.00058

Marshall, P. J., & Stevenson-Hinde, J. (1998). Behavioral inhibition, heart period, and respiratory sinus arrhythmia in young children. *Developmental Psychobiology*, *33*, 283–292. https://doi.org/10.1002/(SICI)1098-2302(199811)33:3%3C283::AID-DEV8%3E3.0.CO;2-N

Edited Books

Marshall, P. J., & Brenneman, K. (Eds.). (2018). *Young children's developing understanding of the biological world*. Routledge. https://doi.org/10.4324/9781315122601

Marshall, P. J., & Fox, N. A. (Eds.). (2006). *The development of social engagement: Neurobiological perspectives*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780195168716.001.0001

Book Chapters

Marshall, P. J. (2018). Embodiment. In A. S. Dick & U. Müller (Eds.), *Advancing developmental science: Philosophy, theory, and method* (pp. 29–40). Routledge. https://doi.org/10.4324/8791315174686

Marshall, P. J. (2015). Neuroscience, embodiment, and development. In W. F. Overton & P. C. M. Molenaar (Eds.), *Handbook of child psychology and developmental science: Vol. 1. Theory and method* (7th ed., pp. 244–283). Wiley. https://doi.org/10.1002/9781118963418.childpsy107

Meltzoff, A. N., Williamson, R. A., & Marshall, P. J. (2013). Developmental perspectives on action science: Lessons from infant imitation and cognitive neuroscience. In W. Prinz, M. Beisert, & A. Herwig (Eds.), *Action science: Foundations of an emerging discipline* (pp. 281–306). MIT Press. https://doi.org/10.7551/mitpress/9780262018555.003.0011

Marshall, P. J., & Fox, N. A. (2007). Electrophysiological measures in research on social and emotional development. In L. A. Schmidt & S. J. Segalowitz (Eds.), *Developmental psychophysiology: Theory, systems, and methods* (pp. 127–149). Cambridge University Press. https://doi.org/10.1017/CBO9780511499791.007

Marshall, P. J., & Fox, N. A. (2007). Infant EEG and ERP in relation to social and emotional development. In M. de Haan (Ed.), *Infant EEG and event-related potentials* (pp. 227–250). Psychology Press. https://doi.org/10.4324/9780203759660-9

Fox, N. A., Schmidt, L. A., Henderson, H. A., & Marshall, P. J. (2007). Developmental psychophysiology: Conceptual and methodological perspectives. In J. T. Cacioppo, L. G. Tassinary, & G. G. Berntson (Eds.), *Handbook of psychophysiology* (3rd ed., pp. 453–481). Cambridge University Press. https://doi.org/10.1017/CBO9780511546396.020

Marshall, P. J., & Fox, N. A. (2006). Biological approaches to the study of social engagement. In P. J. Marshall & N. A. Fox (Eds.), *The development of social engagement: Neurobiological perspectives* (pp. 1–18). Oxford University Press. https://doi.org/10.1093/acprof:oso/9780195168716.003.0001

Fox, N. A., Henderson, H. A., & Marshall, P. J. (2001). The biology of temperament: An integrative approach. In C. A. Nelson & M. Luciana (Eds.), *Handbook of developmental cognitive neuroscience* (pp. 631–645). MIT Press.

Marshall, P. J., & Stevenson-Hinde, J. (2001). Behavioural inhibition: Physiological correlates. In W. R. Crozier & L. E. Alden (Eds.), *International handbook of social anxiety: Concepts, research and interventions relating to the self and shyness* (pp. 53–76). Wiley. Reprinted in W. R. Crozier and L. E. Alden (Eds.), *The essential handbook of social anxiety for clinicians* (2005, pp. 57–80). Wiley.

Marshall, P. J., & Fox, N. A. (2000). Emotion regulation, depression, and hemispheric asymmetry. In S. L. Johnson, A. M. Hayes, T. M. Field, N. Schneiderman, & P. M. McCabe (Eds.), *Stress, coping, and depression* (pp. 35–50). Lawrence Erlbaum Associates. https://doi.org/10.4324/9781410602640

Open Educational Resources

Marshall, P. J. (2026). *About development* (Version 1.0) [Open educational resource]. TUScholarShare. https://doi.org/10.34944/44q3-ss77. Online edition: https://aboutdevelopment.org/

Marshall, P. J. (2026). *Enacting life* (Version 1.0) [Open educational resource]. TUScholarShare. https://doi.org/10.34944/3sta-5a07. Online edition: https://enacting.life/

Paper Presentations and Invited Addresses

Horger, M. N., Campbell, K., & Marshall, P. J. (September, 2026). The impact of fine motor development on infant somatosensory evoked potentials. Paper presented at the annual meeting of the Flux Society, San Diego, CA.

Campbell, K. R., Horger, M. N., Pavlakos, G., Parma, V., & Marshall, P. J. (November, 2025). Concurrent development of neural body maps and grasp control in infancy. Flash talk presented at the annual meeting of the International Society for Developmental Psychobiology, San Diego, CA.

Marshall, P. J. (October, 2025). The space of possibility: Rethinking development for the public good. Presidential address at the biennial meeting of the Society for the Study of Human Development, Lexington, KY.

Marshall, P. J., Campbell, K. R., Pavlakos, G., Horger, M. N., & Parma, V. (May, 2025). Infant grasping and neural body maps: Insights from citizen science and computer vision. Paper presented at the annual convention of the Association for Psychological Science, Washington, DC.

Marshall, P. J. (May, 2024). Studying infant body representations using EEG. Invited symposium presentation at the annual meeting of the Jean Piaget Society, Toronto, ON.

Marshall, P. J. (May, 2024). Probing the development of body maps with infant EEG. Invited symposium presentation at the International Conference on Development and Learning, Austin, TX.

Langer, A., & Marshall, P. J. (April, 2024). Examining teachers' opinions on the potential for using social robots in Philadelphia Public Schools. Paper presented at the annual meeting of the American Educational Research Association, Philadelphia, PA.

Marshall, P. J. (June, 2023). Brains in bodies, bodies in brains. Invited plenary talk at the annual meeting of the Jean Piaget Society, Madrid, Spain.

Campbell, K. R., Horger, M. N., Parma, V., & Marshall, P. J. (March, 2023). Comparing naturalistic and lab measurements: A study of the reach-to-grasp movement. Paper presented at the annual meeting of the Eastern Psychological Association, Boston, MA.

Marshall, P. J. (December, 2021). What role might body maps play in social and cognitive development? Invited presentation at the Center for Mind and Brain, University of California, Davis.

Marshall, P. J. (June, 2021). Embodiment and development. Paper presented at the 50th Anniversary Meeting of the Jean Piaget Society; also served as symposium chair.

Weiss, S. M., & Marshall, P. J. (June, 2021). Comparing child and adult mu rhythm desynchronization in advance of tactile stimulation. Paper presented at the 50th Anniversary Meeting of the Jean Piaget Society.

Weiss, S. M., Laconi, R., & Marshall, P. J. (July, 2019). Individual differences in bodily attention: Anticipatory mu relates to executive function. Paper presented at the annual meeting of the Cognitive Science Society, Montreal, QC.

Marshall, P. J. (June, 2019). The development of neural body representations in infancy. Invited presentation at the International Meeting on Developmental Changes in Pediatric Neurodisability, Groningen, The Netherlands.

Weiss, S. M., Meltzoff, A. N., & Marshall, P. J. (March, 2019). Children's EEG modulation during anticipation of touch: Mu desynchronization relates to executive function and income. Paper presented at the biennial meeting of the Society for Research in Child Development, Baltimore, MD; also served as symposium chair.

Marshall, P. J. (September, 2018). Embodiment and human development. Keynote address at the 8th Joint IEEE International Conference on Development, Learning and Epigenetic Robotics, Tokyo, Japan.

Marshall, P. J. (July, 2018). Connecting self and other in infancy through neural representations of the body. Symposium discussion at the 2018 meeting of the International Congress on Infant Studies, Philadelphia, PA.

Marshall, P. J. (April, 2018). Body maps in the infant brain. Invited presentation at the Department of Psychology, University of Delaware.

Shen, G., & Marshall, P. J. (March, 2018). Exploring categorical and functional boundaries of tactile perception using somatosensory mismatch responses. Invited data blitz presentation at the Annual Meeting of the Cognitive Neuroscience Society, Boston, MA.

Weiss, S. M., & Marshall, P. J. (April, 2017). Anticipatory EEG responses during somatosensory selective attention associated with language and executive function in children. Paper presented at the Biennial Meeting of the Society for Research in Child Development, Austin, TX.

Marshall, P. J. (April, 2017). Embodiment. Invited contributor to Roundtable discussion at the Biennial Meeting of the Society for Research in Child Development, Austin, TX.

Drew, A. R., Meltzoff, A. N., & Marshall, P. J. (April, 2017). Interpersonal influences on body representations in the infant brain. Paper presentation at the Biennial Meeting of the Society for Research in Child Development, Austin, TX.

Marshall, P. J. (June, 2016). Body maps in the infant brain. Invited presentation for the Nurture Science Program, Department of Pediatrics, Columbia University, New York, NY.

Marshall, P. J. (May, 2016). Babies, bodies, and brains. Invited symposium discussion at the Biennial International Congress on Infant Studies, New Orleans, LA.

Marshall, P. J. (May, 2016). Body maps in the infant brain. Preconference presentation at the Biennial International Congress on Infant Studies, New Orleans, LA.

Marshall, P. J. (April, 2016). Body maps in the infant brain. Invited presentation at the Columbia University Seminar in Cognitive and Behavioral Neuroscience, New York, NY.

Marshall, P. J. (September, 2014). The developmental neuroscience of self and other: Connecting infant EEG and MEG. Presentation at the Institute for Learning and Brain Sciences, University of Washington, Seattle, WA.

Marshall, P. J., & Drew, A. R. (March, 2014). A new look at social influences on executive functioning in early childhood. Paper presented at the annual meeting of the Eastern Psychological Association, Boston, MA.

Marshall, P. J. (April, 2013). The future of EEG methods with developing populations. Invited contributor to Roundtable discussion at the Biennial Meeting of the Society for Research in Child Development, Seattle, WA.

Marshall, P. J. (March, 2013). Explanation, mechanism, and development. Distinguished Faculty Lecture at the Center for the Humanities, Temple University.

Marshall, P. J. (June, 2012). Presenting your research: A professor's perspective. Presentation for the McNair Scholars Summer Research Program, Temple University.

Marshall, P. J., Saby, J.N., & Meltzoff, A. N. (June, 2012). Infant EEG responses to action observation vary with expected action outcomes. Paper presented at the International Conference on Infant Studies, Minneapolis, MN.

Saby, J.N., Marshall, P.J., Meltzoff, A.N., & Shipley, T.F. (June 2012). What is reflected by neural mirroring systems? An EEG study of 14-month-olds. Paper presented at the International Conference on Infant Studies, Minneapolis, MN.

Marshall, P. J. (March, 2012). The foundations of social cognition: Current perspectives from developmental psychology and neuroscience. Invited academic presentation for the Distinguished Lecture series, Center for Autism Research, Children's Hospital of Philadelphia, Philadelphia, PA.

Marshall, P. J. (March, 2012). Developing an understanding of others: Current perspectives from developmental psychology and neuroscience. Invited community presentation for the Distinguished Lecture series, Center for Autism Research, Children's Hospital of Philadelphia, Philadelphia, PA.

Marshall, P. J. (January, 2012). Neural mirroring systems in infancy: Current perspectives. Invited brown-bag presentation at the Institute for the Study of Child Development, UMDNJ-Robert Wood Johnson Medical School, New Brunswick, NJ.

Marshall, P. J., & Meltzoff, A. N. (December, 2011). Neural mirroring systems and early learning. Presentation at the NSF Science of Learning Centers Annual Awardee Meeting, Arlington, VA.

Marshall, P. J. (October, 2011). Optimizing class collaboration with wikis. Invited presentation for the Teaching and Learning Center at Temple University.

Marshall, P. J. (October, 2011). Developmental Cognitive Neuroscience: Methods and Applications. Presentation at Grand Rounds, Department of Psychiatry and Behavioral Science, Temple University.

Marshall, P. J. (August, 2011). Exploring the EEG mu rhythm in infancy. Invited presentation given to meeting of P01 Program Project group on Development and Functions of the Mirror Neuron System, College Park, MD.

Marshall, P. J., & Meltzoff, A. N. (March, 2011). Neural mirroring systems: Exploring the EEG mu rhythm in infancy. Paper presentation at the biennial meeting of the Society for Research in Child Development, Montreal, QC.

Saby, J. N., & Marshall, P. J. (March, 2011). EEG methods in the study of early social development: Progress and caveats. Paper presentation at the biennial meeting of the Society for Research in Child Development, Montreal, QC.

Marshall, P. J. (April, 2010). A developmental social neuroscience approach to perception-action relations. Invited talk to the Research in Spatial Cognition group, Department of Psychology, Temple University.

Marshall, P. J. (January, 2010). A developmental social neuroscience approach to perception-action relations. Invited address at the US-Japan Workshop on the Development of the Social Brain, Tokyo, Japan.

Marshall, P. J. (March, 2009). Conceptualizing brain and mind: The view from embodied cognition. Proseminar in the Department of Psychology, Temple University.

Marshall, P. J. (March, 2009). Perception-action interactions: The mirror system and beyond. Brown-bag presentation, Department of Psychology, Temple University.

Marshall, P. J. (October, 2008). Extending the classroom through collaborative projects. Invited presentation to the Instructional Technology User Group at Temple University.

Marshall, P. J. (October, 2008). Relating Psychology and Neuroscience: Taking up the challenges. Brown-bag presentation, Department of Psychology, Temple University.

Marshall, P. J. (September, 2008). Neurobiological perspectives on the effects of early institutional experience: The Bucharest Early Intervention Project. Invited presentation as part of the Neuroscience Program at Temple University.

Marshall, P. J. (April, 2008). A developmental social neuroscience approach to perception-action relations. Invited brown-bag presentation at the Center for Children, Relationships, and Culture, University of Maryland, College Park, MD.

Marshall, P. J. (February, 2008). A developmental social neuroscience approach to perception-action relations. Invited brown-bag presentation at the Institute for the Study of Child Development, UMDNJ-Robert Wood Johnson Medical School, New Brunswick, NJ.

Marx, B., Marshall, P. J, Castro, F. (November, 2007). The moderating effects of stimulus valence and arousal on memory suppression. Paper presented at the 23rd annual meeting of the International Society for Traumatic Stress Studies, Baltimore, MD.

Marshall, P. J., & Fox, N. A. (April, 2007). Electroencephalographic effects of early intervention in institutionalized children. Paper presentation at the biennial meeting of the Society for Research in Child Development, Boston, MA.

Shipley, T. F., Marshall, P. J., & Bouquet, C. (November, 2006). Learning to imitate: EEG analysis of sequential attempts to imitate complex hand movements. Paper presentation at the annual meeting of the Psychonomic Society, Houston, TX.

Marshall, P. J. (September, 2006). Blogs and Wikis: Inspiring and tracking student collaboration using Blackboard. Invited presentation to the Instructional Technology User Group at Temple University.

Marshall, P. J. (September, 2006). Electroencephalographic effects of early intervention in institutionalized children. Invited colloquium at the School of Education, University of Delaware, Newark, DE.

Marshall, P. J., & Hardin, M. G. (April, 2005). Event-related potentials to auditory novelty at 9 months of age in relation to infant temperament. Paper presentation at the biennial meeting of the Society for Research in Child Development, Atlanta, GA.

Marshall, P. J., & Fox, N. A. (April, 2005). EEG power and coherence in children in institutions and in foster care in Romania. Paper presentation at the biennial meeting of the Society for Research in Child Development, Atlanta, GA.

Marshall, P. J. (March, 2004). An analysis of the EEG in institutionalized children in Romania. Invited brown-bag at the Department of Psychology, George Mason University, Fairfax, VA.

Marshall, P. J. (October, 2003). The development of the EEG in institutionalized children in Romania. Paper presentation at the meeting of the American Academy of Child and Adolescent Psychiatry, Miami Beach, FL.

Marshall, P. J. (October, 2003). Temperamental approach and withdrawal tendencies: Recent psychophysiological perspectives. Invited brown-bag at the Department of Psychology, University of Miami, Miami, FL.

Marshall, P. J. (July, 2003). EEG measures of central nervous system development in institutionalized infants and young children in Romania. Invited brown-bag at the Oregon Social Learning Center, Eugene, OR.

Marshall, P. J. (April, 2003). Electrophysiological methods for studying brain-behavior relations. Paper presentation at the biennial meeting of the Society for Research in Child Development, Tampa, FL.

Marshall, P. J., & Fox, N. A. (April, 2003). Electrophysiological measures of auditory novelty processing among institutionalized and community infants and children in Romania. Paper presentation at the biennial meeting of the Society for Research in Child Development, Tampa, FL.

Marshall, P. J., Flanagan, S. P, & Fox, N. A. (October, 2002). ERP and autonomic responses to unattended auditory oddball stimuli during anticipation of an aversive stimulus. Paper presentation at the annual meeting of the Society for Psychophysiological Research, Washington, DC.

Marshall, P. J., & Fox, N. A. (April, 2002). The development of the EEG in institutionalized infants and young children in Romania. Paper presentation at the biennial meeting of the International Society on Infant Studies, Toronto, ON.

Marshall, P. J. (October, 2001). Development of EEG power spectra from 5 months to 4 years of age. Paper presentation at the annual meeting of the Society for Psychophysiological Research, Montreal, QC.

Bar-Haim, Y., & Marshall, P. J., (April, 2001). Auditory sensory processing in socially withdrawn children. Paper presentation at the biennial meeting of the Society for Research in Child Development, Minneapolis, MN.

Marshall, P. J., & Burgess, K. B. (April, 2001). Do attachment and temperament predict behavioral adjustment and physiology at age 4 years? Paper presentation at the biennial meeting of the Society for Research in Child Development, Minneapolis, MN.

Marshall, P. J., & Fox, N. A. (September, 2000). Central and peripheral correlates of approach and withdrawal tendencies across childhood. Invited presentation at the Occasional Temperament Conference, Mystic, CT.

Henderson, H. A., & Marshall, P. J., (April, 1999). Behavioral inhibition and frontal asymmetry: Relations to social development in middle childhood. Paper presentation at the biennial meeting of the Society for Research in Child Development, Albuquerque, NM.

Marshall, P. J. (October, 1998) Attention and vagal regulation in 2-year-olds. Paper presentation at the annual meeting of the Society for Psychophysiological Research, Denver, CO.

Marshall, P. J. (July, 1997). The Madingley studies of behavioural inhibition. The selected sample at 4.5 and 7 years: Cardiac reactivity. Paper presentation at the International Conference on Self-Consciousness and Shyness, University of Wales at Cardiff, UK.

Poster Presentations

Eulau (Campbell), K., Dib, S., Horger, M. N., Parma, V., & Marshall, P. J. (September, 2026). A window into the developing brain-body connection: Body maps and temperament in infancy. Poster presented at the annual meeting of the Flux Society, San Diego, CA.

Marshall, P. J., & Campbell, K. R. (June 19, 2026). The space of possibility: Explaining reliability and variability in development. Poster presented at the 52nd Annual Meeting of the Society for Philosophy and Psychology, Baltimore, MD.

Campbell, K. R., Horger, M. N., Pavlakos, G., Parma, V., & Marshall, P. J. (October, 2025). Exploring the developmental possibility space: Variability in infant prehension. Poster presented at the biennial meeting of the Society for the Study of Human Development, Lexington, KY.

Marshall, P. J., Jaeger, W., & Dib, S. (May, 2025). Teaching development through a broader biological lens: An open educational resource for psychological science. Poster presented at the annual meeting of the Association for Psychological Science, Washington, DC.

Campbell, K. R., Eulau, K., Weiss, S. M., & Marshall, P. J. (May, 2025). Tactile attention in childhood: Relations with executive functioning and temperament. Poster presented at the biennial meeting of the Society for Research in Child Development, Minneapolis, MN.

Howard, L. H., Wilson, J. R., Langer, A., & Marshall, P. J. (August, 2024). Draw a robot task: The influence of children's age and drawings on robot interactions. Late-breaking report presented at the 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Pasadena, CA.

Campbell, K. R., Parma, V., & Marshall, P. J. (May, 2024). More, more, more data! Quantitatively deriving infant grasping strategies. Poster presented at the IEEE International Conference on Development and Learning, Austin, TX.

Campbell, K. R., Eulau, K., Parma, V., & Marshall, P. J. (March, 2024). Investigating the neural correlates of grasp development in infancy. Poster presented at the annual meeting of the Eastern Psychological Association, Philadelphia, PA.

Dib, S., Campbell, K. R., Parma, V., & Marshall, P. J. (March, 2024). Relations between infant negative affectivity and neural indices of grasp development. Poster presented at the annual meeting of the Eastern Psychological Association, Philadelphia, PA.

Scheri, N., Marshall, P. J., Campbell, K. R., & Horger, M. N. (March, 2024). Predicting infant fine motor skills at 14 months from early grasping kinematics. Poster presented at the annual meeting of the Eastern Psychological Association, Philadelphia, PA.

Campbell, K. R., Horger, M. N., Parma, V., Scheri, N., Dib, S., & Marshall, P. J. (October, 2023). Motor development and neural correlates of behavior. Poster presented at the biennial meeting of the Society for the Study of Human Development, Philadelphia, PA.

Campbell, K. R., Parma, V., & Marshall, P. J. (September, 2023). Neural correlates of tactile attention in infancy. Poster presented at the annual meeting of the Fetal, Infant, & Toddler Neuroimaging Group, Santa Rosa, CA.

Campbell, K. R., Eulau, K., Weiss, S. M., & Marshall, P. J. (September, 2023). Ignore the tap: Neural correlates of children's voluntary tactile attention. Poster presented at the annual meeting of the Flux Society, Santa Rosa, CA.

Campbell, K. R., Horger, M. N., Parma, V., & Marshall, P. J. (November, 2022). Sex differences in the maturation of the grasp in infancy: A longitudinal kinematic analysis. Poster and poster-pitch presentation at the annual meeting of the International Society for Developmental Psychobiology, San Diego, CA.

Kelly, R., Mochnal, L. E., Hill, K., Campbell, K. R., Horger, M. N., Marshall, P. J., & Parma, V. (June, 2022). Developmental changes in infant reach-to-grasp behavior: A kinematic analysis of caregiver cellphone videos. Poster presented at the annual meeting of the Jean Piaget Society, Philadelphia, PA.

Campbell, K. R., Eulau, K., Horger, M. N., & Marshall, P. J. (June, 2022). The impact of routines on parent-reported temperament: Associations between surgency and reach speed. Poster presented at the annual meeting of the Jean Piaget Society, Philadelphia, PA.

Campbell, K., Weiss, S. M., & Marshall, P. J. (November, 2021). Investigating the ability to shift attention between interoceptive and exteroceptive stimulation. Poster presented at the annual meeting of the International Society for Developmental Psychobiology. Online.

Boyle, B., Weiss, S. M., & Marshall, P. J. (April, 2021). Relation between parent and child screen time and COVID-related screen time is mediated by executive function. Poster presented at the biennial meeting of the Society for Research in Child Development, online.

Rodriguez, Y., Weiss, S. M., & Marshall, P. J. (March, 2019). Executive function and language abilities differentially relate to indicators of socioeconomic status. Poster presented at the Annual Meeting of the Eastern Psychological Association, New York, NY.

Weiss, S. M., Meltzoff A. N., & Marshall, P. J. (August, 2018). Anticipating touch: Mu desynchronization during bodily attention relates to executive function in children. Poster presented at Flux, the Annual Meeting of the Society for Developmental Cognitive Neuroscience, Berlin, Germany.

Weiss, S. M., Smyk, N. & Marshall, P. J. (April, 2018). Comparing brain responses during anticipation of tactile stimulation initiated by a robot or a human. Poster presented at the 2018 APA Technology, Mind, and Society Conference, Washington, DC.

Taylor, J. R., Smyk, N., Weiss, S. M., & Marshall, P. J. (March, 2018). Sensations and actions in social robots: Anthropomorphization predicts likability. Poster presented at the 2018 Annual Meeting of the Eastern Psychological Association, Philadelphia, PA.

Smyk, N., Weiss, S. M., Taylor, J. R., & Marshall, P. J. (March, 2018) Comparing sensorimotor oscillations during a motor task with a robot or human partner. Poster presented at the 2018 Annual Meeting of the Cognitive Neuroscience Society, Boston, MA.

Weiss, S. M., Laconi, R., & Marshall, P. J. (March, 2018). Anticipatory EEG activity during somatosensory selective attention relates with executive function. Poster presented at the 2018 Annual Meeting of the Cognitive Neuroscience Society, Boston, MA.

Weiss, S. M. & Marshall, P. J. (July, 2017) Children's EEG indices of directed attention during somatosensory anticipation: Relations with executive function. Poster presented at the annual meeting of the Cognitive Science Society, London, UK.

Weiss, S. M., & Marshall, P. J. (April, 2017). Anticipation of touch influences children's EEG responses during somatosensory selective attention. Poster presented at the biennial meeting of the Society for Research in Child Development, Austin, TX.

Shen, G., & Marshall, P. J. (November, 2016). Effects of auditory processing on tactile sensory evoked potentials. Poster presented at the annual meeting of the Society for Neuroscience, San Diego, CA.

Weiss, S. M., & Marshall, P. J. (August, 2016). Parsing selective attention and executive function in children. Poster presented at the annual meeting of the Cognitive Science Society, Philadelphia, PA.

Shen, G., Saby, J. N., & Marshall, P. J. (April, 2016). Exploring social influences on brain potentials during anticipation of tactile stimulation. Poster presented at the annual meeting of the Cognitive Neuroscience Society, New York, NY.

Drew, A. R., Meltzoff, A. N., & Marshall, P. J. (April, 2016). Exploring individual differences in attentional modulation of the somatosensory evoked potential. Poster presented at the annual meeting of the Cognitive Neuroscience Society, New York, NY.

Drew, A. R., & Marshall, P. J. (March, 2015). Exploring social influences on anticipatory EEG activity in a cued spatial attention task. Presented at the annual meeting of the Cognitive Neuroscience Society, San Francisco, CA.

Saby, J. N., Meltzoff, A. N., & Marshall, P. J. (March, 2015). Neural correlates of the infant body schema: Scalp topography of ERPs to tactile stimulation in 7-month-olds. Poster presented at the biennial meeting of the Society for Research in Child Development, Philadelphia, PA.

Drew, A. R., & Marshall, P. J. (April, 2014). Influence of perspective and hand used on sensorimotor EEG responses during action observation. Presented at the annual meeting of the Cognitive Neuroscience Society, Boston, MA.

Drew, A. R., Quandt., L. C., & Marshall, P. J., (April, 2013). Expectations about action outcome modulate ERP responses to observed actions. Presented at the annual meeting of the Cognitive Neuroscience Society, San Francisco, CA.

Marshall, P. J., & Saby, J. N., (April, 2013). Determinants of response inhibition in preschoolers: Why is the Simon Says task so difficult for young children? Poster presented at the biennial meeting of the Society for Research in Child Development, Seattle, WA.

Saby, J. N., Meltzoff, A. N., & Marshall, P. J. (April, 2013). Neural measures of imitative learning: Somatotopic mu rhythm desynchronization over sensorimotor cortex in 14-month-old infants. Poster presented at the biennial meeting of the Society for Research in Child Development, Seattle, WA.

Saby, J. N., Ridge, K., Meltzoff, A. N., & Marshall, P. J. (June, 2012). Delineating EEG responses to different types of actions at 14 months of age. Poster presented at the International Conference on Infant Studies, Minneapolis, MN.

Quandt, L. C., Marshall, P. J., & Shipley, T. F. (September, 2011). EEG responses to observed gestures vary with somatosensory expectations. Poster presented at the annual meeting of the Society for Psychophysiological Research, Boston, MA.

Saby, J. N., Sartor, C. E., Meltzoff, A. N., & Marshall, P. J. (March, 2011). Cross-modal repetition suppression of the infant mu rhythm response to action observation: New directions. Poster presented at the biennial meeting of the Society for Research in Child Development, Montreal, QC.

Saby, J. N., Smythe, R., Sartor, C. E., & Marshall, P. J. (March, 2011). Effects of perceived agency and biological versus non-biological motion on motor contagion in four-year-old children. Poster presented at the biennial meeting of the Society for Research in Child Development, Montreal, QC.

Quandt, L. C., Marshall, P. J., & Shipley, T. F. (April, 2011). Is the EEG mu rhythm sensitive to the predicted outcomes of observed actions? Presented at the annual meeting of the Cognitive Neuroscience Society, San Francisco, CA.

Quandt, L. C., Marshall, P. J., Bouquet, C. A., Young, T., & Shipley, T. F. (September, 2010). The effect of visual and motor experience during early stages of imitative learning. Poster presented at the annual meeting of the Society for Psychophysiological Research, Portland, OR.

Marshall, P. J., Meltzoff, A. N., Sartor, C. E., & Young, T. (March, 2010). EEG desynchronization during action observation and action execution at 14 months of age. Poster presented at the biennial meeting of the International Society on Infant Studies, Baltimore, MD.

Marshall, P. J., McCann, S. M., & Meltzoff, A. N. (March, 2010). Self-Perception: Responses to temporal and spatial manipulations in displays of self-produced movement at 5 Months. Poster presented at the biennial meeting of the International Society on Infant Studies, Baltimore, MD.

Marshall, P. J., Schimeneck, B. J., & Meltzoff, A. N. (March, 2010). Infant looking behavior during maternal interactions at 5 Months: Double video vs. face-to-face. Poster presented at the biennial meeting of the International Society on Infant Studies, Baltimore, MD.

Vanderwert, R. E., Reeb-Sutherland, B. C., Marshall, P. J., Fox, N. A. (March, 2010). Early intervention effects on the developing EEG: Recent findings from the Bucharest Early Intervention Project (BEIP). Poster presented at the biennial meeting of the International Society on Infant Studies, Baltimore, MD.

Funk, A. Y., Sartor, C. E., & Marshall, P. J. (May, 2009). The development of children's conceptualizations of brain and mind: Implications for education. Poster presented at the biennial conference of the International Mind, Brain, and Education Society, Philadelphia, PA.

Schimeneck, B. J., McCann, S. M., LaFrance, L. N., Sartor, C. E., Meltzoff, A. N., & Marshall, P. J. (April, 2009). Behavioral reactions of 5-month-olds to live versus delayed video images of their mothers. Poster presented at the biennial meeting of the Society for Research in Child Development, Denver, CO.

McCann, S. M., Schimeneck, B. J., LaFrance, L. N., Sartor, C. E., Meltzoff, A. N., & Marshall, P. J. (April, 2009). Perception of the bodily self in 5-month-old infants: Manipulation of viewpoints and temporal delay. Poster presented at the biennial meeting of the Society for Research in Child Development, Denver, CO.

Thomas, A. L., Marshall, P. J., Bouquet, C. A., & Shipley, T. F. (April, 2009). Studying motor contagion in four-year-old children. Poster presented at the biennial meeting of the Society for Research in Child Development, Denver, CO.

Castro, F., Sandt, A. R., & Marshall, P. J. (October, 2008). Postauricular reflex potentiation during safe/threat associative learning. Poster presented at the annual meeting of the Society for Psychophysiological Research, Austin, TX.

Marshall, P. J., Shipley, T. F., Fisher, K., LaFrance, L., & Catella, S. J. (March, 2008). Event-related EEG mu rhythm responses to point-light actions in 5-month-old infants. Poster presented at the biennial meeting of the International Society on Infant Studies, Vancouver, BC.

Reeb, B. C., Marshall, P. J., & Fox, N. A. (March, 2008). Temperamental differences in electrophysiological responses to auditory novelty in 3-year-old children. Poster presented at the biennial meeting of the International Society on Infant Studies, Vancouver, BC.

Shipley, T. F., Marshall, P. J., Bouquet, C. (August, 2007). Perception-action coupling in learning new skills: EEG analysis of imitation of hand movements. Poster presented at the European Conference on Visual Perception, Arezzo, Italy.

Catella, S. J., LaFrance, L., Burden, L. M., Shipley, T. F., & Marshall, P. J. (July, 2007). Behavioral and neurophysiological assessment of perception-action interactions in preschool children. Invited poster presented at NSF/Autism Speaks workshop on Neurocognitive Development and Autism, Arlington, VA.

Fisher, K. R., Nanayakkara, A., & Marshall, P. J. (May, 2007). Neurophysiological and behavioral correlates of error monitoring and student motivation. Poster presented at the meeting of the Jean Piaget Society, Amsterdam, The Netherlands.

Fitzhugh, S. L., Shipley, T. F., & Marshall, P. J. (May, 2007). The relation between motor cortex activity and perception of form coherence for biological motion stimuli. Poster presented at the meeting of the Vision Sciences Society, Sarasota, FL.

Fisher, K. R., Marshall, P.J., & Shipley, T. F. (March, 2007). Activation of the infant mirror neuron system during observation of point-light walkers. Poster presented at the biennial meeting of the Society for Research in Child Development, Boston, MA.

Catella, S. J., LaFrance, L., Burden, L. M., Shipley, T. F., & Marshall, P. J. (March, 2007). Behavioral and neurophysiological assessment of perception-action interactions in preschool children. Poster presented at the biennial meeting of the Society for Research in Child Development, Boston, MA.

Castro, F., Marx, B. P., Marshall, P. J., & Soler-Baillo, J. M. (November, 2006). The moderating effect of emotion in suppressing unwanted memories. Poster presented at the annual convention of the Association for Behavioral & Cognitive Therapies, Chicago, IL.

Hardin, M., Marshall, P. J., & Fox, N. A. (May, 2004). Electrophysiological indices of auditory novelty processing in temperamentally selected 9-month-old infants. Poster presented at the International Conference on Infant Studies, Chicago, IL.

Lozano, D. I., Raine, A., Baker, L. A., Bezdjian, S., & Marshall, P. J. (October, 2002). Reduced mismatch negativity in violent children. Poster presented at the annual meeting of the Society for Psychophysiological Research, Washington, DC.

Marshall, P. J., Bar-Haim, Y., & Schorr, E. (October, 2001). Mismatch negativity to tones and speech stimuli in middle childhood. Poster presented at the annual meeting of the Society for Psychophysiological Research, Montreal, Canada.

Marshall, P. J., Bar-Haim, Y., & Fox, N. A. (October, 2000). Mismatch negativity and P50 gating in middle childhood. Poster presented at the annual meeting of the Society for Psychophysiological Research, San Diego.

Himmelfarb, D., Marshall, P. J., Bar-Haim, Y., & Fox, N. A. (July, 2000). Relations of early negative reactivity to later social behavior, internalizing problems, and cardiac patterns: Effects of stimulus modality. Poster presented at the biennial meeting of the International Society on Infant Studies, Brighton, UK.

Marshall, P. J., & Fox, N. A. (October, 1999). Cortical influences on high-frequency heart period variability. Poster presented at the annual meeting of the Society for Psychophysiological Research, Granada, Spain.

Marshall, P. J., & Stevenson-Hinde, J. (April, 1995). Behavioural inhibition and heart period in young children. Poster presented at the biennial meeting of the Society for Research in Child Development, Indianapolis, IN.

Grants and Awards

Computational phenotyping of early motor skill
University of Washington
Role: PI
Project period: 06/26 – 05/28
$25,000 direct costs

Expanding the horizons for research at the developmental interface of brain and behavior
James S. McDonnell Foundation
Role: PI
Project period: 01/21 – 07/24
$250,000 direct costs

Development and neural plasticity of body representations
University of Washington
Role: Contract PI
Project period: 06/20 – 06/24
$100,000 direct costs

Children's selective attention and brain responses to tactile stimulation
PA Department of Health
Role: PI
Project period: 07/17 – 12/20
$100,000 direct costs

Body maps in early infancy: A foundation for social engagement
NSF BCS-1460889
Role: PI
Project period: 07/15 – 06/19 including no-cost extension
$191,498 direct costs, $287,618 total costs.

Development of neural body maps
NSF SMA-1540619
Role: Co-PI
Project period: 09/15 – 08/18
$170,065 direct costs, $265,301 total costs

Neuroscience of self-other correspondence in infancy
NIH (NICHD) 1R21HD083756-01A1
Role: PI
Project period: 08/15 – 07/18 including no-cost extension
$203,579 direct costs, $291,736 total costs.

Social and cognitive influences on action processing in infancy
University of Washington
I-LABS Research Innovation & Visiting Scholar Funding
Project period: 07/13 – 08/15
$229,605 direct costs

Infants processing of actions and goals: A social neuroscience approach
NIH (NICHD) 1R21HD068734
Role: PI
Project period: 07/11 – 06/14 including no-cost extension
$275,000 direct costs, $420,750 total costs

The neuroscience of action processing in infancy
University of Washington
I-LABS Research Innovation Funding
Project period: 10/10 – 09/11
$30,000 direct costs

A developmental social neuroscience approach to perception-action relations
NSF BCS-0642404
Role: PI
Project period: 08/07 – 07/11 including no-cost extension
$402,150 direct costs, $577,035 total costs (includes two supplements)

Mastery motivation and error monitoring in gifted children: Behavioral and neurophysiological correlates
American Psychological Foundation
Role: PI
Project period: 09/05 – 06/07
$25,000 direct costs

Electrophysiological correlates of action perception and imitation in children
PA Department of Health Tobacco Formula Funding: Individual Investigator Award
Role: PI
Project period: 01/06 – 06/07
$50,000 direct costs, $58,145 total costs

Electrophysiology and temperament in institutionalized children
MacArthur Network on Early Experience and Brain Development Junior Faculty Award
Role: PI
Project period: 03/01 – 03/04
$104,000 direct costs

Incentive motivation, coping style, and EEG asymmetry
NIH (NIMH) 1R03MH059043
Role: PI
Project period: 09/98 – 08/00
$25,000 direct costs, $37,500 total costs

Internal Support from Temple University

2027	Sabbatical (Spring semester)
2024	Textbook Affordability Project award (TU Libraries)
2023	Grant in Aid
2014	Sabbatical (Academic Year)
2007	Study Leave (Fall semester)
2007	OVPR and CLA Research Incentive Fund awards
2006	Summer Research Award

Awards

2013	Fellow, Association for Psychological Science
2012	Lindback Award for Distinguished Teaching at Temple University
2012	Faculty Fellow, Center for the Humanities at Temple University

Departmental, College, and University Service at Temple University

2016–26	Chair, Department of Psychology & Neuroscience
2023–25	Research Council, Office of the Vice President for Research
2020–24	Strategic Planning Steering Committee, Provost's Office
2020–23	Strategic Planning Steering Committee, College of Liberal Arts
2017–18	Academic Technology Committee, College of Liberal Arts

2016–19	Dean's Research Council, College of Liberal Arts
2016–18	Tenure Track Merit Committee, College of Liberal Arts
2016–17	Research and Service Awards Committee, College of Liberal Arts
2016–17	Dean's Strategic Planning Steering Committee, College of Liberal Arts
2016–17	Infrastructure Working Group (Chair), College of Liberal Arts
2016–17	Dean's Ad Hoc Committee on TU Japan, College of Liberal Arts
2015–16	Neuroscience Search Committee, Department of Psychology
2013–14	Web & Technology Committee, Department of Psychology
2011–12	Cognitive Search Committee, Department of Psychology
2010–16	Developmental Area Director, Department of Psychology
2010–14	Representative Faculty Senator, College of Liberal Arts
2010–11	Neuroscience Search Committee, Department of Psychology
2010–16	Graduate Committee, Department of Psychology
2009–12	Faculty Mentor for Psi Chi, International Honors Society in Psychology
2008–09	Neuroscience Search Committee, Department of Psychology
2006–13	Undergraduate Committee, Department of Psychology
2006–07	Cognitive Neuroscience Search Committee, Department of Psychology
2004–05	Cognitive Neuroscience Search Committee, Department of Psychology

Editorial Responsibilities

2014–20	Associate Editor, *Developmental Science*
2014–19	Associate Editor, *International Journal of Behavioral Development*
2015–16	Guest Editor, *Early Education and Development*
2010–	Editorial Board, *Development and Psychopathology*
2003–13	Editorial Board, *Infancy*
2005–06	Editorial Board, *SRCD Monographs*

Professional Service and Leadership

2025–26	COGDOP/ORCA working group on promotion and tenure guidelines
2023–25	President, Society for the Study of Human Development
2021–23	President-Elect, Society for the Study of Human Development
2018–21	Board of Directors, Council of Graduate Departments of Psychology (Treasurer role)
2017–19	Reviewer, NIH Special Emphasis Study Section on Risk Prevention and Health Behavior (RPHB).
2011–15	Appointed Member, NIH Study Section on Psychosocial Development, Risk and Prevention (PDRP).
2012	Chair, NIH Special Emphasis Study Section on Risk Prevention and Health Behavior (RPHB).
2010	Reviewer, NICHD Human-Animal Interaction Study Section.
2010	Reviewer, NIH Special Emphasis Study Section on Risk Prevention and Health Behavior (RPHB).
2010	Reviewer, NIH Study Section on Psychosocial Development, Risk and Prevention (PDRP).
2009	Reviewer, Institute of Education Sciences (IES) review panel for Early Intervention and Early Childhood Education.
2004–	Reviewer of grant proposals for various funding agencies, including Academy of Finland, Vienna Science and Technology Fund, NWO (Netherlands), EU and ANR (France), Science Foundation Ireland, BBSRC and ERSC (UK), Netherlands Organisation for Scientific Research.
2004–	Ad-hoc book proposal and book chapter reviews for various publishers including Cambridge University Press, Oxford University Press, and Princeton University Press.